exerciciosR4.R

r3914162

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```r
# 1)
v1 = c(2,5,2,6,1,4)
A = matrix(v1,2,3,1)
A
```

```
##      [,1] [,2] [,3]
## [1,]    2    5    2
## [2,]    6    1    4
```

```r
A = A*3

# 2)
D = matrix(c(15,18,21,27,18,10,14,5,4),3,3,1)
D
```

```
##      [,1] [,2] [,3]
## [1,]   15   18   21
## [2,]   27   18   10
## [3,]   14    5    4
```

```r
D*2
```

```
##      [,1] [,2] [,3]
## [1,]   30   36   42
## [2,]   54   36   20
## [3,]   28   10    8
```

```r
# 3)
v2 = matrix(c(1,2,2,4,7,6),3,2,1)
v3 = matrix(c(10,20,30,40,50,60),3,2,1)
B = 2/7*(v2-v3)

# 4)
A = B%*%A

# 5)
range(A)
```

```
## [1] -351.4286  -54.0000
```

```r
# 6)
sum(A)
```

```
## [1] -1635.429
```

```r
# 7)
prod(A)
```

```
## [1] -5.689671e+19
```

```r
# 8)
sum(A[, ])
```

```
## [1] -1635.429
```

```r
# 9)
sum(B[B<4])
```

```
## [1] -53.71429
```

```r
#10)
library(readxl)
Exercicios <- read_excel("Exercicios.xlsx")
Exercicios
```

```
## # A tibble: 6 x 4
##   Nome           Idade Sexo     NF
##   <chr>          <dbl> <chr> <dbl>
## 1 José Santos       17 M        92
## 2 Angela Dias       17 F        75
## 3 Aline Souza       16 F        81
## 4 Mayara Costa      15 F        87
## 5 Lara Lins         15 F        90
## 6 Nicolas Barros    13 M        88
```

```r
is.data.frame(Exercicios)
```

```
## [1] TRUE
```

```r
Exercicios$Sexo <- as.factor(Exercicios$Sexo)
is.factor(Exercicios$Sexo)
```

```
## [1] TRUE
```

```r
mean(Exercicios$Idade)
```

```
## [1] 15.5
```